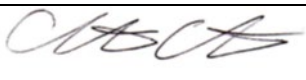


Ventricular Assist Devices (VADs)

Guideline Title: Ventricular Assist Devices (VADs)

Guideline Number: GUID-3203-PR031

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- I. **PURPOSE:** To maximize the quality of care, and improve short and long-term outcomes, for the patient requiring ventricular support systems during both scene & interfacility transport.

II. **ABOUT Ventricular Assist Devices (VADs):**

A VAD is a surgically implanted mechanical pump that is attached to the heart that works with the heart to help it pump more blood with less work. It does this by continuously taking blood from the left ventricle and moving it to the aorta, which then delivers oxygen-rich blood throughout the body. The VAD has both internal and external components. The actual pump sits on or next to the heart's left ventricle with a tube attached that routes the blood to the aorta. A cable called a drive line extends from the pump, out through the skin, and connects the pump to a controller and power sources worn outside the body. The drive line must be connected to the controller, and the controller must be connected to power at all times to keep the pump working properly. The pump is powered by batteries or electricity. Some VADs have an adaptor that also allows them to run off a car battery. Each device has specific carrying cases to allow the patient to move about freely with the equipment.

Since the devices are continuous flow the patient may present without palpable pulses, no blood pressure and extremely low pulse oximetry since it is measured by pulsatile flow. Most VAD patients have a medical alert bracelet or a color-coded tag on themselves or their equipment that uniquely identifies them. The color-coded tag will have what type of device it is, what hospital placed the device and a contact number to the VAD coordinator. The current VADs in use include the **Heartware - HVAD**, the **Heartmate II**, and the **Heartmate 3 – MVAD**.

III. **DEFINITIONS:**

- a. ACT – Activated clotting time

- b. AICD - Automatic Implantable Cardioverter Defibrillator
- c. CO – Cardiac output
- d. MAP – Mean arterial pressure
- e. VAD - Ventricular assist device
- f. PCU - Patient Control Unit

IV. **EQUIPMENT**

- a. Tubing clamps
- b. Patient Control Unit (PCU)
- c. Equipment: patient's primary console, back-up controller, at least two back-up batteries, battery clips, universal battery charger, vest, and the mobile power unit.

V. **DEPARTMENT GUIDELINE:**

- a. Establish care as defined by *GUID-G002 – General Patient Care*.
- b. Complete a physical exam of patient to include
 - i. Assess the system components
 - 1. Check all lead connections.
 - 2. Reconnect any loose or disconnected leads.
 - 3. Auscultate Heart Sounds to determine if the device is functioning and what type of device it is. If it is continuous flow device, you should hear a "humming sound." This indicates the device has power.
 - 4. Assess for a green circular light. This indicates the device has power and is functioning. A black circle indicates the device is inoperable.
 - 5. The toggle switch allows view of 4 parameters: Sped (4,000-6,200 rpm), Flow (LPM), Power (w) and Pulsatility Index (PI).
 - ii. Assess and manage airway, breathing and circulation.
 - 1. Provide supplemental oxygen. Pulse oximetry is inaccurate because it is continuous flow. Use skin color, capillary refill, and skin temperature.
 - 2. Pulses will be weak to absent in most VAD patients.
 - 3. Obtain a MAP using a manual BP cuff. Utilize a doppler if available (interfacility). The first swoosh sound is the MAP. Compare this to the noninvasive measurement. The most accurate measure is the doppler, BUT this assessment is not possible during air transport.
 - a. If pulsatile: Goal 65-85 mmHg
 - b. If nonpulsatile: Goal 60-80 mmHg
 - iii. Assess for any alarms and fix. See below for Alarm Management.
- c. The patient is the expert on the device. Please use them for guidance if mental status appropriate.
- d. A medical alert bracelet or a color-coded tag on the patient's controller near their waist will direct you to the type of the device the patient has.
- e. Use the EMS guide for a reference for emergency operations (below).
- f. During air transport
 - i. Non-invasive blood pressure monitor - Goal for MAP 65-90 mmHg.

- ii. Arterial Lines – Use MAP. Goal for MAP: 65-80 mmHg. Waveform may appear flat.
- g. Notify the VAD coordinator and provide them with your assessment, rhythm, vital signs, alarms, and performance data from the pocket controller (rpms, watts, flow, and PI). This can be obtained by pressing the toggle switch on the distal end of the controller.
 - i. VAD coordinator should be notified for all orders regarding the VAD and guidance on any alarms you are unable to troubleshoot.
 - ii. Communicate capillary refill & skin color to the VAD coordinator.
 - iii. All orders given by the VAD coordinator are supported by the Medical Director.
 - iv. Central Florida Area VAD Coordinators (24-7 contact information)
 - 1. AdventHealth Orlando:
 - a. Advanced Heart Failure Team PH: 407-303-2474
 - 2. Orlando Regional Medical Center:
 - a. Mechanical Circulatory Support (MCS) PH: 689-837-2443
 - b. VAD/MCS Coordinator is available by Perfect Serve under Mechanical Circulatory Support
 - c. AHF Physicians and APPs can be reached via Perfect Serve
 - 3. Mayo Clinic Jacksonville Florida:
 - a. (Pending confirmation) Call main number and ask for SICU charge nurse
 - b. PH: 904-953-2000
 - 4. Shands Hospital (UFHealth) Gainesville:
 - a. Call ShandsCair Dispatch and ask for online medical control who will transfer you to one of their ED doctors for medical direction
 - b. PH: 800-342-5365
 - 5. St. Joseph's Hospital – BayCareTampa
 - a. Call BayCare Transfer center and ask for on call VAD nurse
 - b. PH: 727-561-2449. Press option 2 to speak to a nurse.
 - 6. Tampa General Hospital Ventricular Assist Device Program 1-800-505-7769
- h. Discuss with the VAD coordinator the most appropriate receiving facility. If patient has an immediate life threat, transport to the closest appropriate facility for stabilization. If patient appears stable and the problem isn't with the device, transport to the facility that placed the VAD.
 - i. NOTE: Orlando Regional does NOT have Heartware Medtronic batteries or back up equipment. Transport to AdventOrlando or

contact a National Medtronic coordinator (PH: 877-367-4823 M-F 7a-6p).

- i. Transport the significant other if space is available and the pilot/team approves.
- j. **Take the patient's equipment (travel bag) or any VAD equipment with them.** It is vitally important to have back up equipment while in transport.
- k. Start an IV of NS 0.9% or Lactated Ringers (LR) at KVO.
- l. Treatment considerations
 - i. If the patient experiences ventricular dysrhythmias requiring cardioversion/defibrillation proceed with treatment per cardiac guidelines and per *GUID-PR004 – Cardioversion & GUID-PR0008 Defibrillation*. DO NOT DISCONNECT THE CONTROLLER when defibrillating or during cardioversion. Many patients have an AICD.
 - ii. Cardiac arrest can still occur in the setting of a functioning VAD but may be more difficult to identify. If patient experiences cardiopulmonary arrest, manual or mechanical compressions are permitted. Since compressions may cause the cannula to dislodge, quickly evaluate other potential causes of unresponsiveness:
 - 1. Rule out other causes of unconsciousness (hypoglycemia, medication side effects, hypovolemia/hemorrhage, etc.)
 - 2. Check VAD for alarms and assess for functionality (auscultating for hum when possible)
 - 3. If patient has no palpable pulse and there is no arterial line, use of continuous capnography is an excellent measure to assure perfusion (ETCO₂ maintained above 20 mmHg is suggestive of continued perfusion).
 - 4. Assess Hs & Ts.
 - 5. Notify Air Care Dispatch to page the VAD surgeon immediately
- m. Alarm Management – Red (Life Threat & intervene immediately) or Yellow (Advisory)
 - i. Low Battery Alarm - Change to fresh batteries or to AC power.
 - 1. Batteries last 8 - 12 hours.
 - 2. The system controller of the HeartMate 3 has a 15-minute battery. You cannot silence that alarm
 - 3. Note HeartMate 3 batteries work with the HeartMate 2 device
 - ii. Low Flow Alarm (< 2.5 L/min). Assess for poor filling of the pump or obstruction to emptying. Utilize the SHAB DORC Acronym.
 - 1. **S**epsis or **S**edation - low flow and low MAP < 70 mmHg
 - a. Consider IV fluid bolus 250 ml NS 0.9% or LR and continue treatment per *GUID-M011 – Hypotension*.
 - b. Assess for drive line infection
 - 2. **H**ypertension - MAP > 90 mmHg affects the outflow causing a low flow state. If MAP > 90 mmHg recheck BP in 10 minutes the SVR may be too elevated:

- a. Treat by contacting the VAD coordinator. Consider vasodilator.
- 3. **Arrhythmias** - If ventricular tachycardia is symptomatic treat per *GUID-C006 - Narrow and Wide Complex Tachycardia*
 - a. Suction event - presents with occasional flashes RED HEART alarm and/or runs of ventricular ectopy.
 - b. Consider IV fluid bolus 250 ml NS 0.9% or LR.
- 4. **Bleeding:**
 - a. Assess for potential GI bleeding
 - b. If actively hemorrhaging with hypotension, control the bleeding and consider infusing blood products per *GUID-PR002 - Blood Product Administration* or *GUID-PR025 – Emergency Blood Product Administration*.
 - c. TXA is not contraindicated for VAD patients with life threatening hemorrhage.
- 5. **Dehydration:** Consider IV fluid bolus 250 ml NS 0.9% or LR.
- 6. **Overdrive:**
 - a. Speed may be too fast for the patient's current volume status.
 - b. If obstruction suspected, treat by changing the controller to the spare located in the patient's travel bag.
- 7. **Right Side Heart Failure:** Contact the VAD coordinator. Consider inotropes (Dopamine, Dobutamine, Epinephrine infusion) or pacing to maintain preload.
- 8. **Clot:**
 - a. The watts on the controller will be gradually increasing due to thrombosis.
 - b. Contact the VAD coordinator for instruction.
 - c. Treatment is a complete VAD change (surgery) or anticoagulation- all directed by the VAD coordinator.
- 9. Consider all Hs & Ts
- n. **Special Considerations**
 - i. If the device has been off for 5 minutes do not restart, UNLESS instructed by the VAD coordinator.
 - ii. The system controller does get hot. Patients should not keep controller on their abdomen as it can burn the skin.
- o. **Medical Information**
 - i. Patients should be on Aspirin & Anticoagulation. If on Coumadin, goal is INR 2-3.
 - 1. If INR is high, do not administer Vitamin K. Consider Plasma or let it trend down.
 - ii. LDH levels are markers for thrombosis
 - a. Assess patients for signs of complications: drive line or other infection, thrombosis, stroke, GI bleeding,

and right ventricular failure due to pulmonary hypertension or pulmonary embolism.

- iii. Treat the patient, not the pump.
- iv. Patients are unable to have an MRI
- p. Additional Helpful Information
 - i. At Orlando Regional
 - 1. VAD equipment is kept in: Cardiac Surgery, CVICU, 7B, 7N and the VAD Clinic.

VI. **DOCUMENTATION: EMS Charts**

- a. Document contact and direction from VAD coordinator.
- b. Document MAP and other vital signs.
- c. Document green circular light on controller.
- d. Document method in which MAP obtain.
- e. Document speed (RPM), flow (LPM), pulsatility index (PI) and power (watts).
- f. Document any alarms and action required.
- g. Type and model of device
- h. Condition of driveline and skin insertion site

VII. **REFERENCES:**

- a. EMS Field Guides – MyLVAD <https://www.mylvad.com/medical-professionals/resource-library/ems-field-guides>
- b. Heartmate 3 LVAD
<https://www.cardiovascular.abbott/us/en/hcp/products/heart-failure/left-ventricular-assist-devices/heartmate-3/about.html>
- c. Mechanical Circulatory Support Emergency Guide
https://www.mylvad.com/sites/default/files/EMS%20Field%20Guides/2021%20Field%20Guides/ICCAC_Emergency_Guides_20_21-1.pdf
- d. Moazami, N. (2012). Patients with a Ventricular Assist Device Need Special Considerations. *Journal of Emergency Medical Services*.
- e. Rakesh, G., et al. (2019). Device Therapy and Arrhythmia Management in Left Ventricular Assist Device Recipients: A Scientific Statement From the American Heart Association. *Circulation*, e967–e989.
<https://doi.org/10.1161/CIR.0000000000000673>
- f. Ozdemir Koken, Z., Sezer, R.E., & Kervan, U. (2019, September). Caring for Patients With Ventricular Assist Devices: A Mini-Review of the Literature. *Transplantation Proceedings*, 51(7), 2492-2494.
<https://doi.org/10.1016/j.transproceed.2019.03.046>

The QR Code will give you the Field Guide of each Vendor

